Phase 2 Integrated Clustering Report

Credit Card Transaction Clustering

This report compares partitioning, hierarchical, density-based, and model-based clustering algorithms. Model selection uses only internal metrics. Fraud labels are used only for post-hoc external evaluation.

The recommended Phase 2 clustering is Hierarchical with parameters k=2,linkage=single. The recommended number of clusters is k=2. This recommendation was determined only from internal clustering metrics. The selected configuration obtained Silhouette=0.88923, Davies-Bouldin=0.07741, and Calinski-Harabasz=132.71336. It achieved the strongest aggregate rank across Silhouette, Davies-Bouldin, and Calinski-Harabasz among the final candidate configurations. Its Silhouette coefficient was 0.889233, indicating strong geometric separation. Its bootstrap membership stability was classified as moderate.

Reasons:
- It achieved the strongest aggregate rank across Silhouette, Davies-Bouldin, and Calinski-Harabasz among the final candidate configurations.
- Its Silhouette coefficient was 0.889233, indicating strong geometric separation.
- Its bootstrap membership stability was classified as moderate.

Limitations:
- The external fraud label was not used for model selection; external scores are post-hoc descriptive evidence only.
- The fraud class is highly imbalanced, so purity can be misleading.
- A transaction profile cluster is not the same thing as a fraud class.
- Sensitivity to changing Euclidean distance to Manhattan distance was classified as low.
- Among the inspected lowest-silhouette records, the negative-silhouette fraction was 0.000000.
- Hierarchical partitions depend on linkage choice and dendrogram cutting strategy.

Final Algorithm Comparison

algorithm	parameters	n_clusters	noise_fraction	silhouette	davies_bouldin	calinski_harabasz	ari	nmi	ami	purity	runtime_seconds
KMeans	k=2	2.0000	0.0000	0.1807	2.2213	284.3639	0.0037	0.0005	-0.0001	0.9980	0.1626
Hierarchical	k=2,linkage=single	2.0000	0.0000	0.8892	0.0774	132.7134	-0.0003	0.0000	-0.0003	0.9980	0.3540
DBSCAN	eps=11.405267,min_samples=30	1.0000	0.0042				0.2671	0.1594	0.1584	0.9980	0.5918
GMM	k=8,covariance=full	8.0000	0.0000	0.0362	4.1473	134.7120	0.0014	0.0051	0.0044	0.9980	1.1759

Internal Ranking Scoreboard

final_order	algorithm	parameters	selected_k	silhouette	davies_bouldin	calinski_harabasz	seed_ari_mean	internal_rank_sum
1.0000	Hierarchical	k=2,linkage=single	2.0000	0.8892	0.0774	132.7134		5.0000
2.0000	KMeans	k=2	2.0000	0.1807	2.2213	284.3639	0.2967	5.0000
3.0000	GMM	k=8,covariance=full	8.0000	0.0362	4.1473	134.7120	0.5557	8.0000

K-Means Search

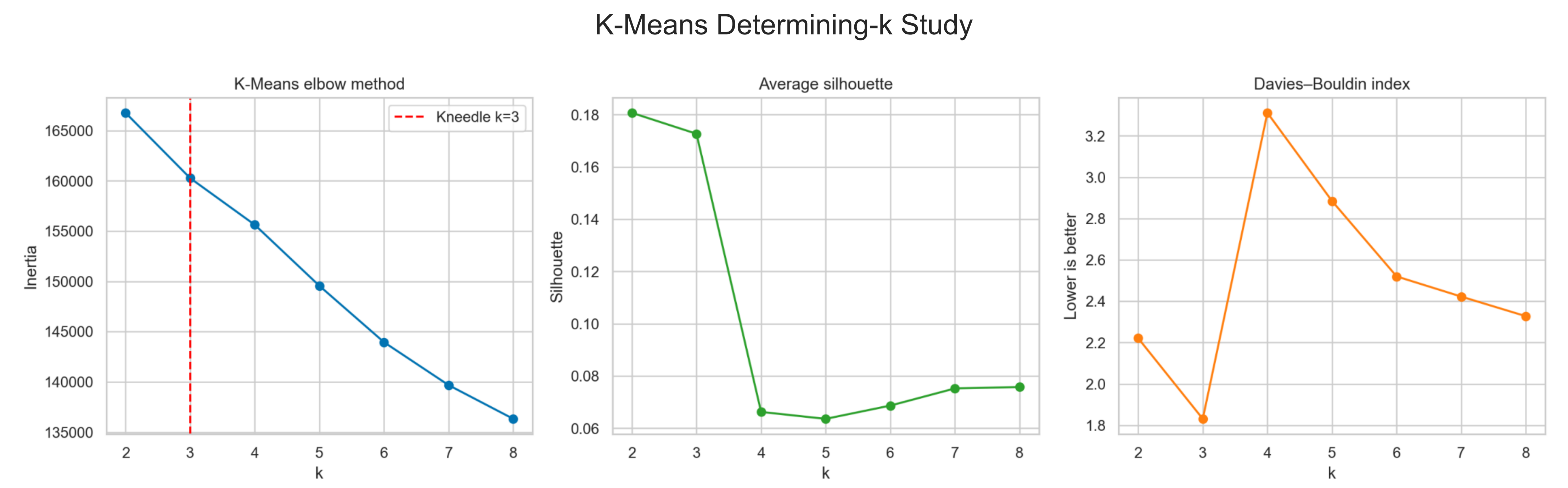
k	inertia	silhouette	davies_bouldin	calinski_harabasz	runtime_seconds
2.0000	166769.4433	0.1807	2.2213	284.3639	0.4332
3.0000	160269.1000	0.1727	1.8311	269.5400	0.2848
4.0000	155624.5130	0.0663	3.3115	244.6753	0.2369
5.0000	149556.3288	0.0636	2.8849	251.7359	0.2700
6.0000	143946.8951	0.0687	2.5201	255.9141	0.2648
7.0000	139675.3762	0.0753	2.4226	250.2929	0.3191
8.0000	136322.6382	0.0758	2.3277	240.8293	0.3308

Bootstrap Stability

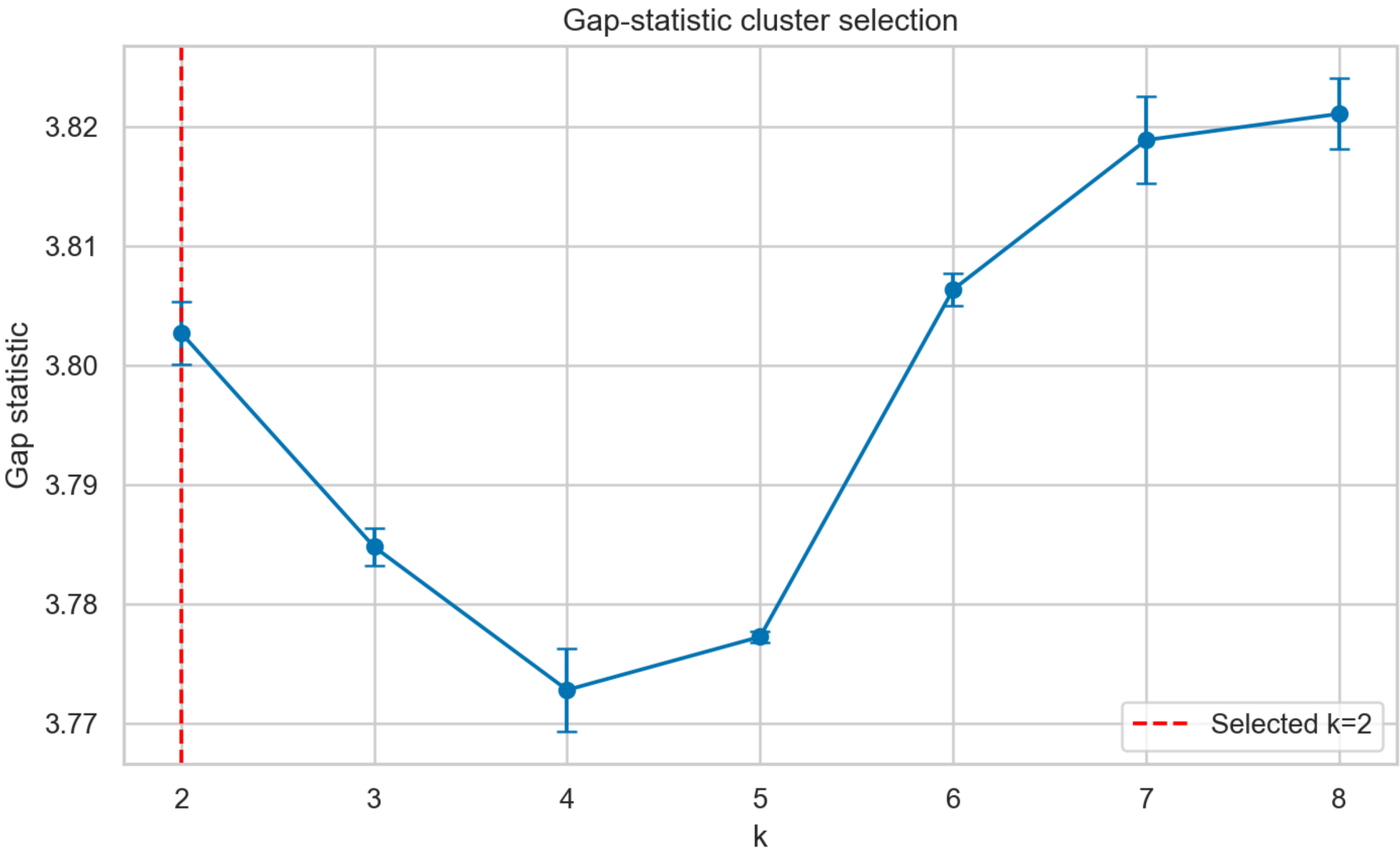
k	bootstrap_runs	ari_mean	ari_std	mean_jaccard	jaccard_std	worst_cluster_mean_jaccard
2.0000	20.0000	0.3555	0.4853	0.6677	0.2522	0.4041
3.0000	20.0000	0.4636	0.4503	0.5763	0.2234	0.3019
4.0000	20.0000	0.5021	0.2883	0.6392	0.1501	0.2926
5.0000	20.0000	0.3482	0.1779	0.6536	0.0911	0.3928
6.0000	20.0000	0.6441	0.2908	0.6827	0.1199	0.0909
7.0000	20.0000	0.6192	0.2190	0.6617	0.1050	0.1221
8.0000	20.0000	0.6149	0.1428	0.6381	0.0961	0.0835

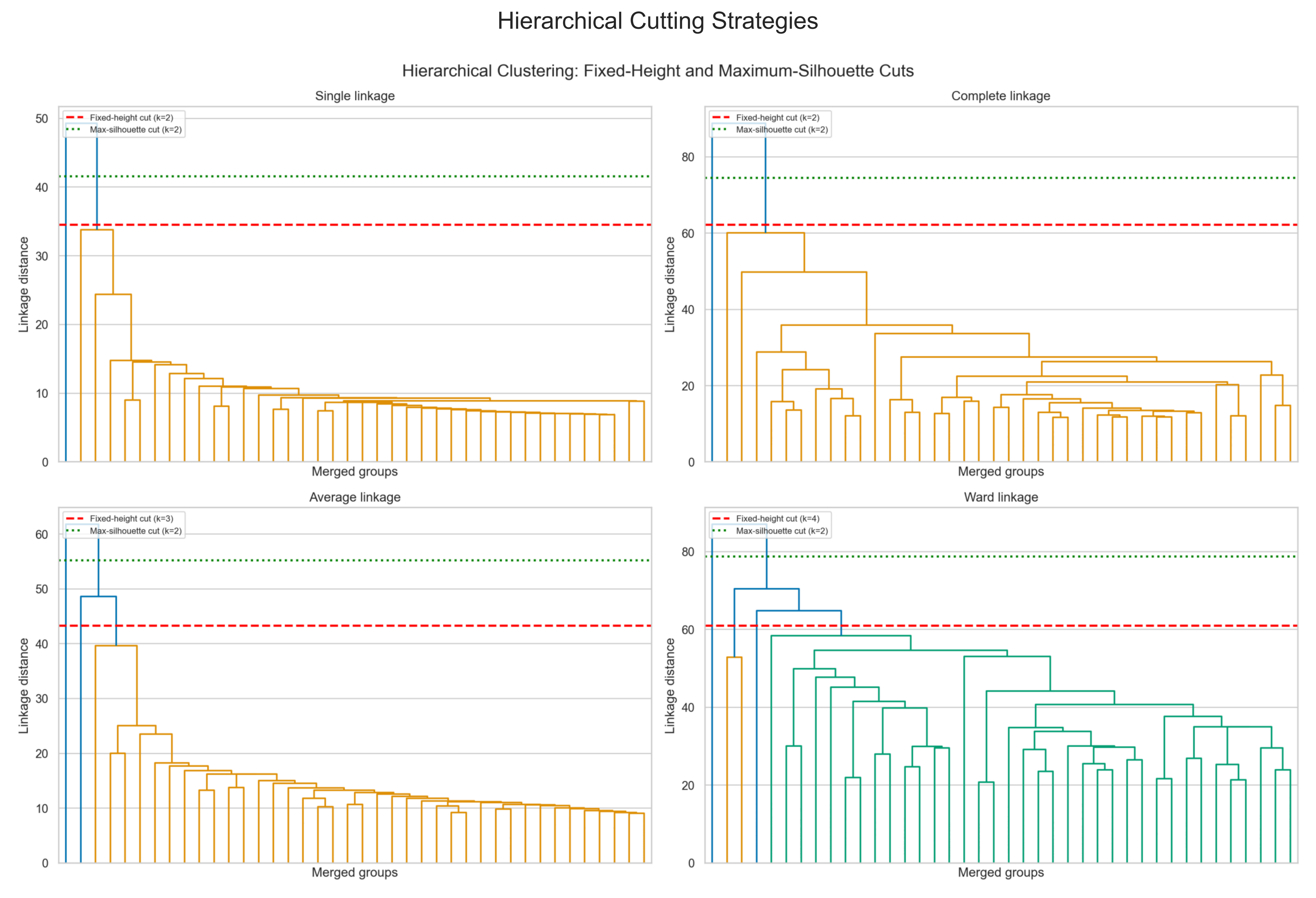
Distance-Metric Sensitivity

distance_metric	requested_k	actual_k	silhouette_matching_metric	davies_bouldin	calinski_harabasz	ari_external	runtime_seconds
euclidean	2.0000	2.0000	0.8881	0.0773	131.0199	-0.0007	0.0971
manhattan	2.0000	2.0000	0.8896	0.0773	131.0199	-0.0007	0.0895


K-Means Determining-k Study
K-Means elbow method
Kneedle k=3
Inertia
165000
160000
155000
150000
145000
140000
135000
2
3
4
5
6
7
8
k
Average silhouette
Silhouette
0.18
0.16
0.14
0.12
0.10
0.08
0.06
Davies–Bouldin index
Lower is better
3.2
3.0
2.8
2.6
2.4
2.2
2.0
1.8

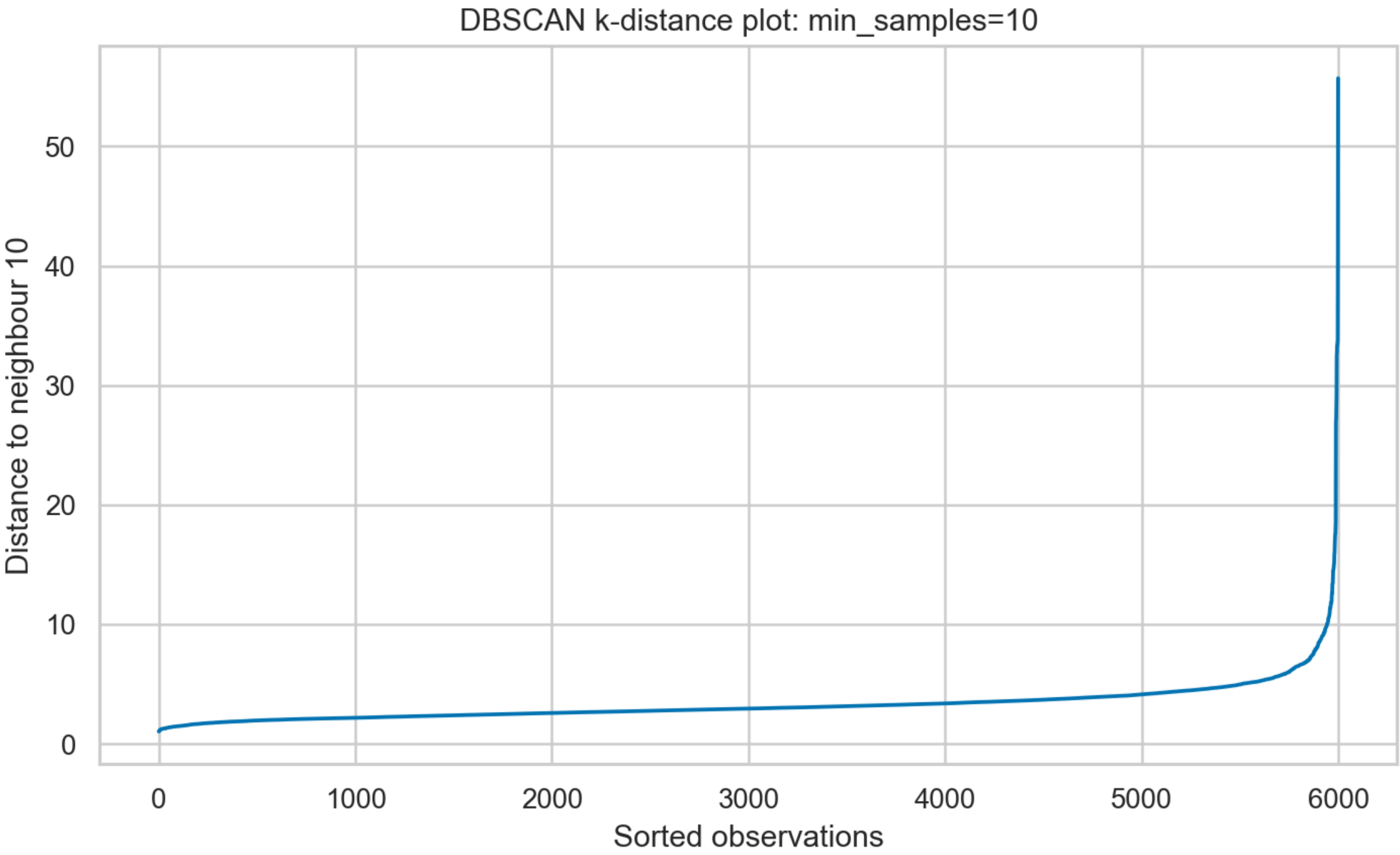
Gap Statistic

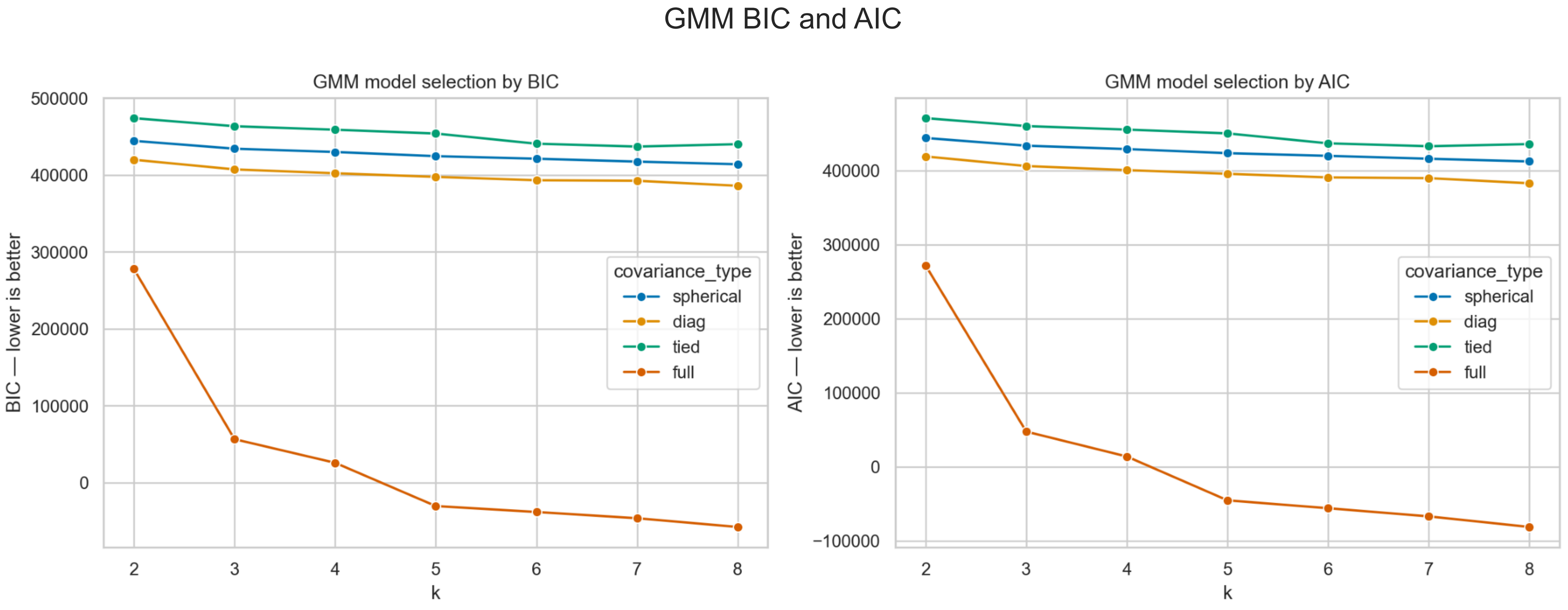


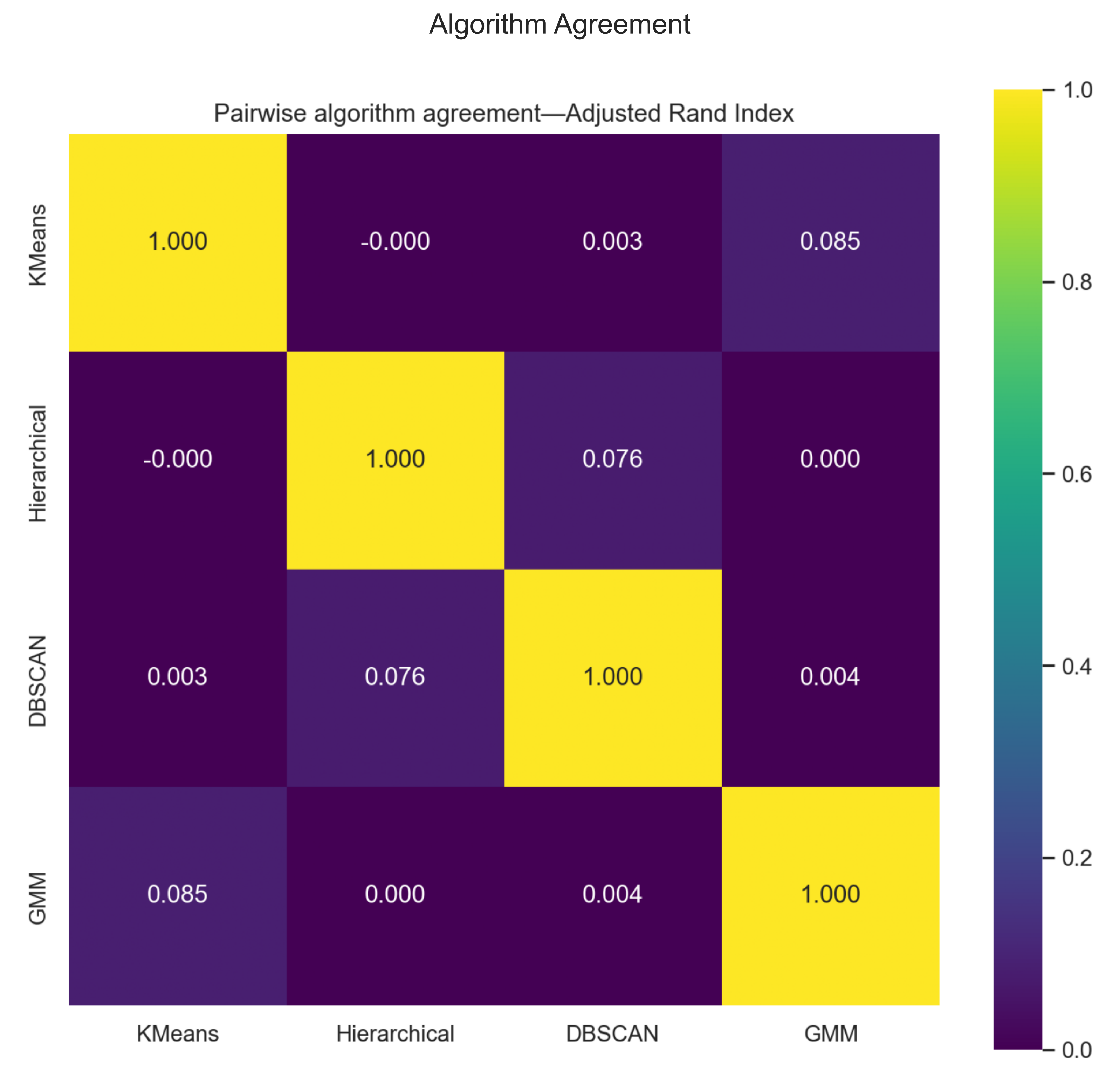


Hierarchical Cutting Strategies
Hierarchical Clustering: Fixed-Height and Maximum-Silhouette Cuts
Single linkage
Fixed-height cut (k=2)
Max-silhouette cut (k=2)
Linkage distance
Merged groups
Complete linkage
Fixed-height cut (k=2)
Max-silhouette cut (k=2)
Linkage distance
Merged groups
Average linkage
Fixed-height cut (k=3)
Max-silhouette cut (k=2)
Linkage distance
Merged groups
Ward linkage
Fixed-height cut (k=4)
Max-silhouette cut (k=2)
Linkage distance
Merged groups

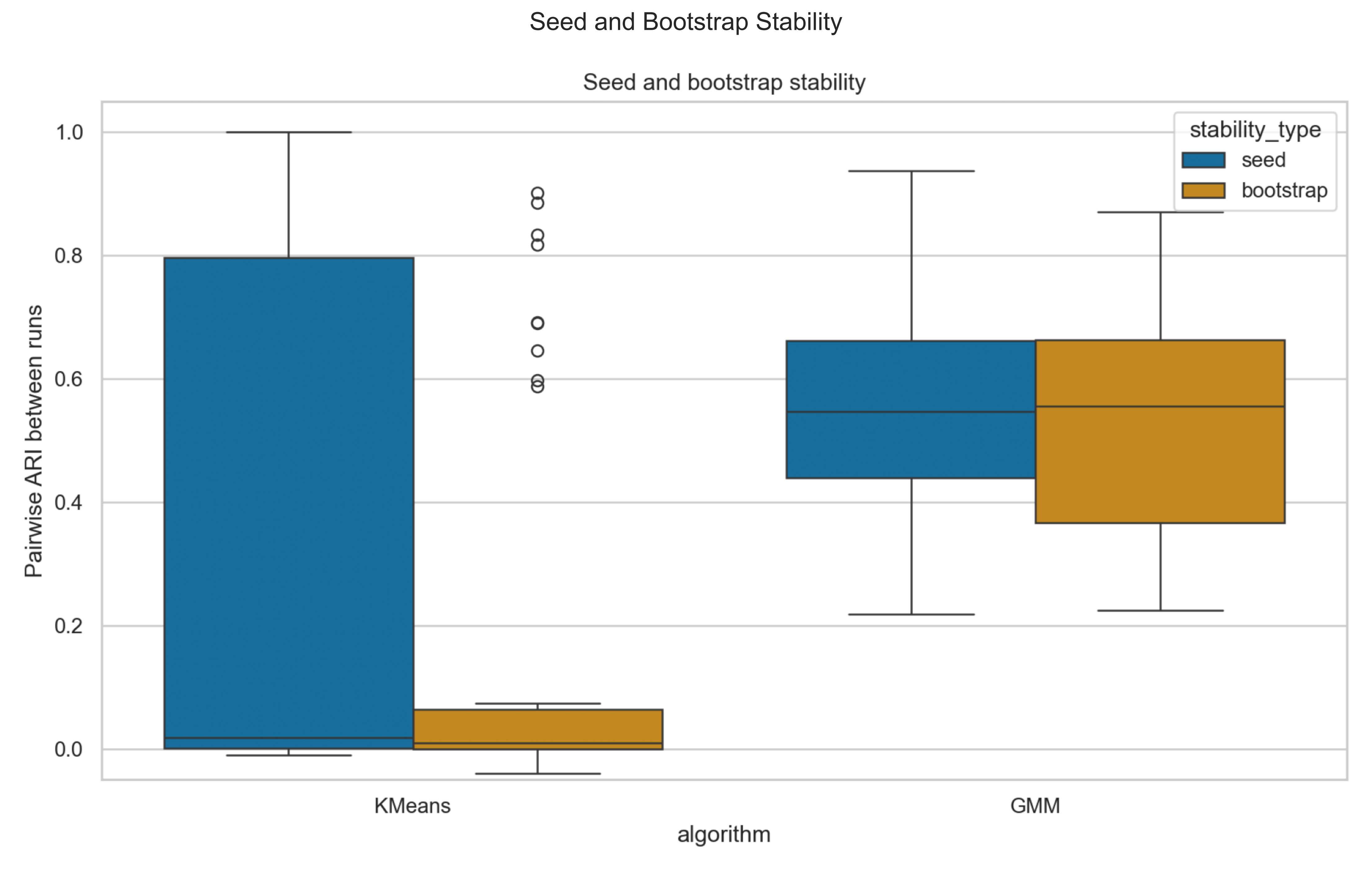
DBSCAN k-Distance Diagnostic

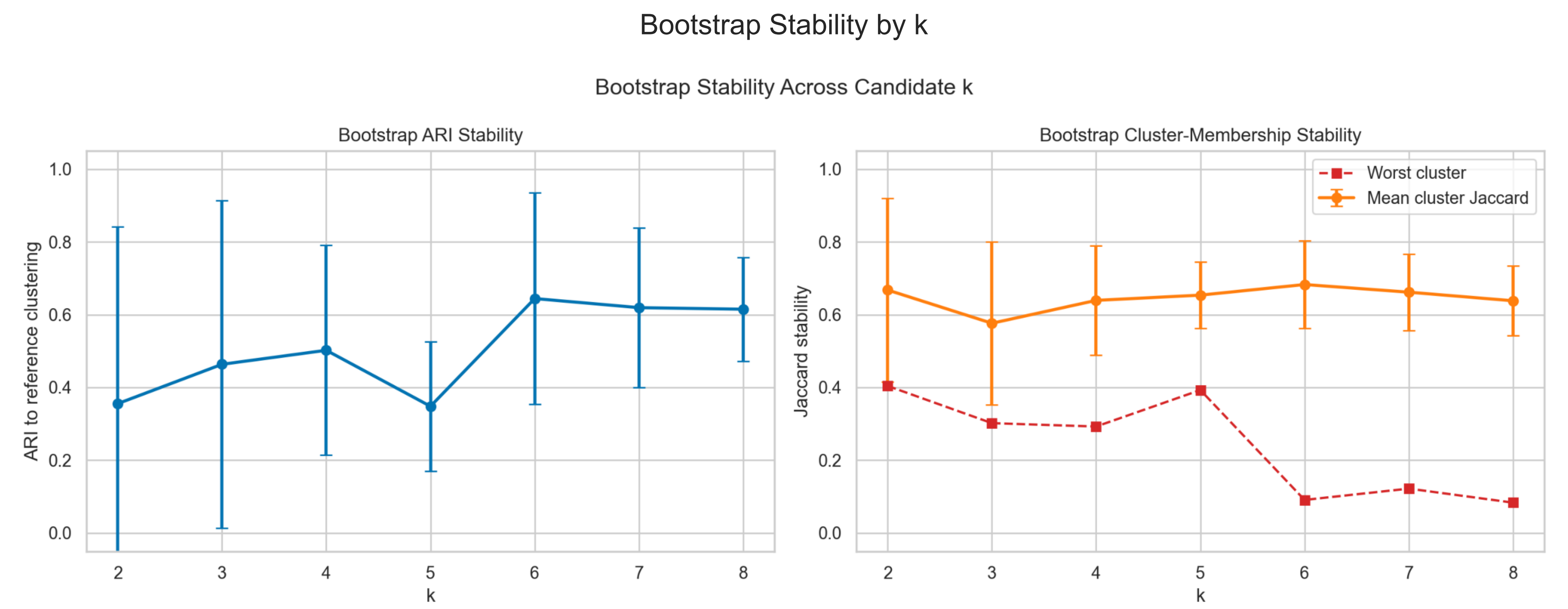



GMM BIC and AIC
GMM model selection by BIC
BIC — lower is better
500000
400000
300000
200000
100000
0
2
3
4
5
6
7
8
k
covariance_type
spherical
diag
tied
full
GMM model selection by AIC
AIC — lower is better
400000
300000
200000
100000
0
−100000



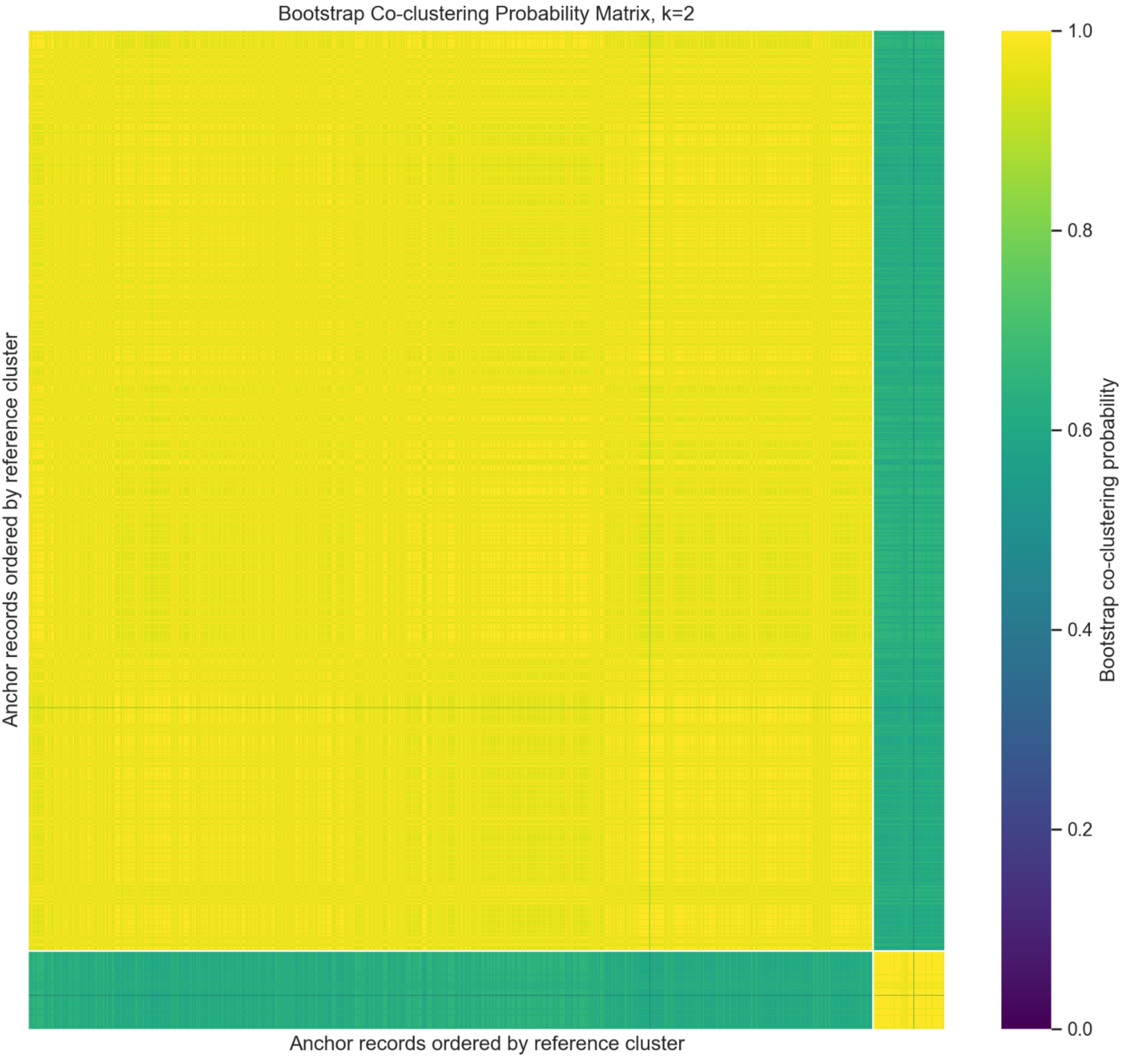
Algorithm Agreement
Pairwise algorithm agreement—Adjusted Rand Index
KMeans
Hierarchical
DBSCAN
GMM
1.000
-0.000
0.003
0.085
-0.000
1.000
0.076
0.000
0.003
0.076
1.000
0.004
0.085
0.000
0.004
1.000
1.0
0.8
0.6
0.4
0.2
0.0

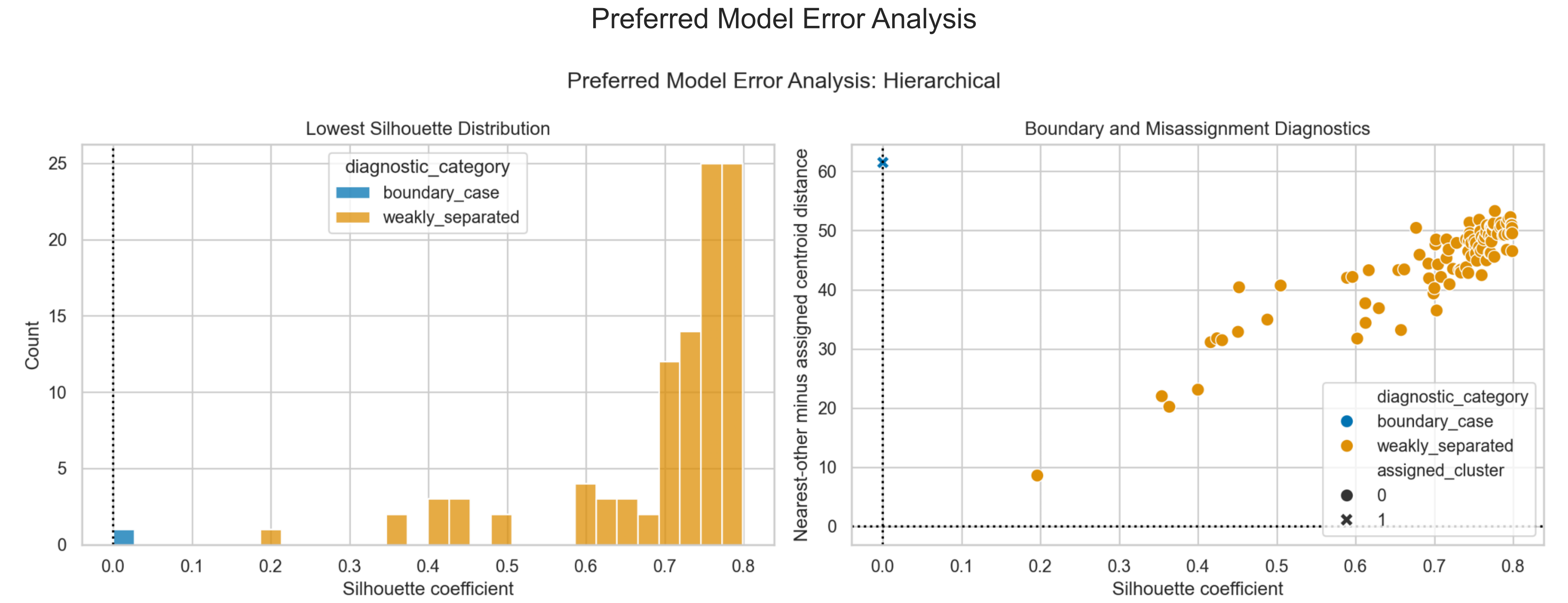

Seed and Bootstrap Stability
Seed and bootstrap stability
stability_type
seed
bootstrap
Pairwise ARI between runs
1.0
0.8
0.6
0.4
0.2
0.0
KMeans
GMM
algorithm



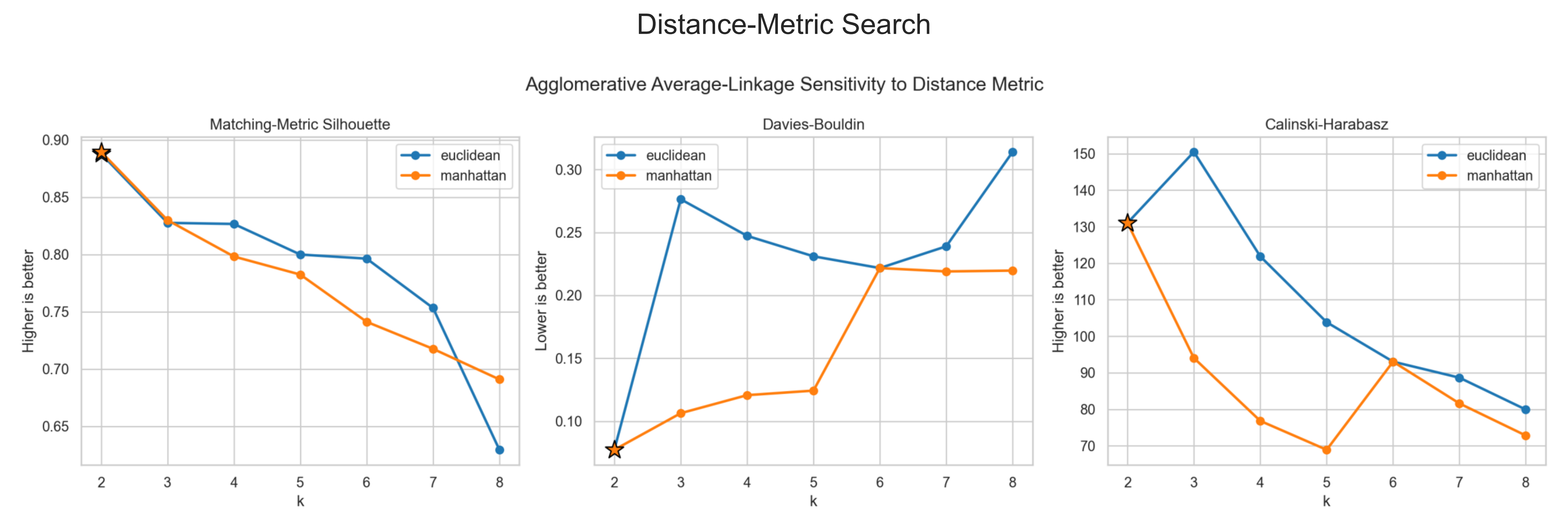
Bootstrap Stability by k
Bootstrap Stability Across Candidate k
Bootstrap ARI Stability
ARI to reference clustering
k
Bootstrap Cluster-Membership Stability
Jaccard stability
Worst cluster
Mean cluster Jaccard

Bootstrap Co-clustering Matrix

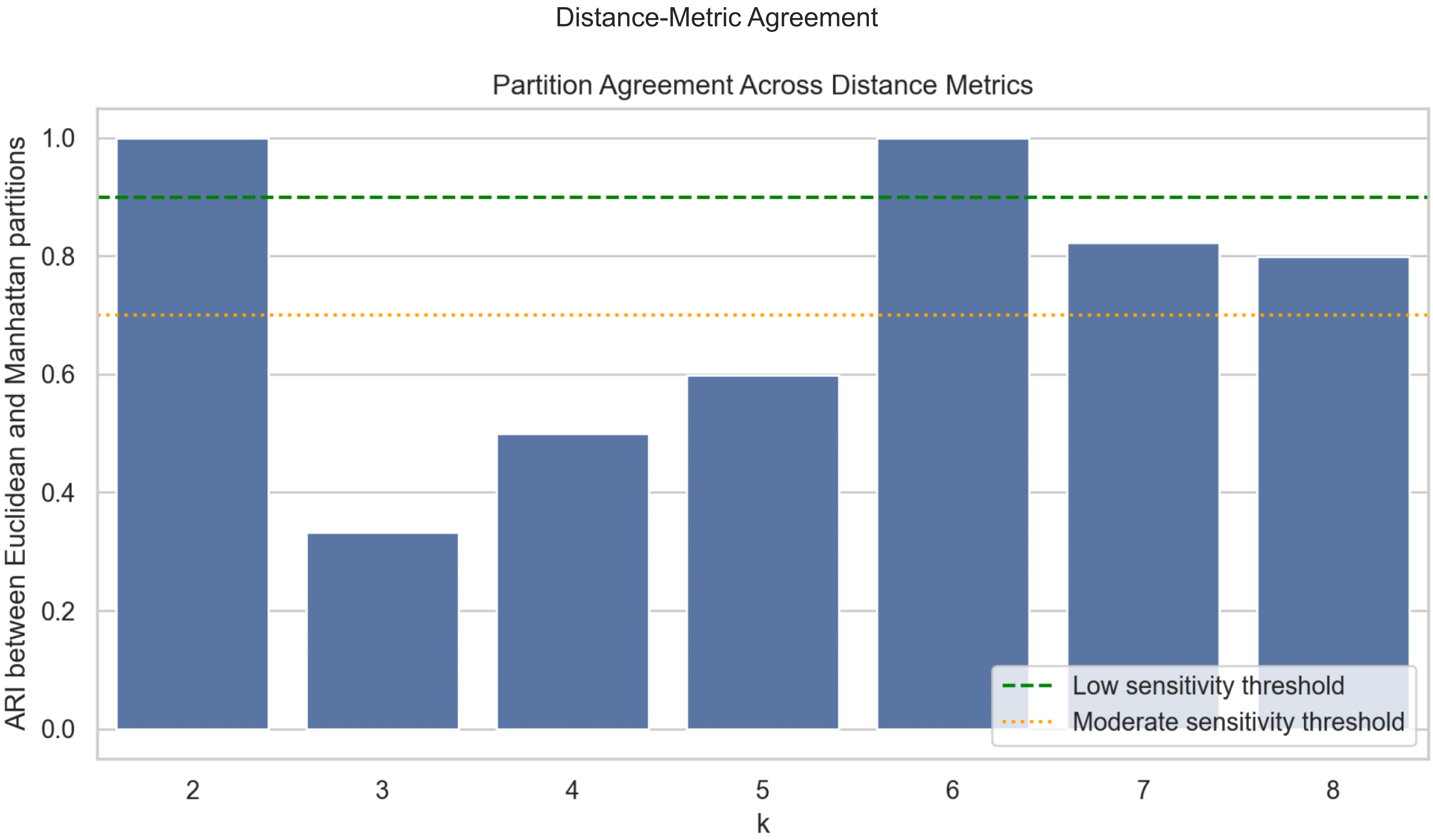




Preferred Model Error Analysis
Preferred Model Error Analysis: Hierarchical
Lowest Silhouette Distribution
diagnostic_category
boundary_case
weakly_separated
Count
0
5
10
15
20
25
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Silhouette coefficient
Boundary and Misassignment Diagnostics
Nearest-other minus assigned centroid distance
0
10
20
30
40
50
60
diagnostic_category
boundary_case
weakly_separated
assigned_cluster
0
1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Silhouette coefficient



Distance-Metric Search
Agglomerative Average-Linkage Sensitivity to Distance Metric
Matching-Metric Silhouette
Higher is better
euclidean
manhattan
k
Davies-Bouldin
Lower is better
euclidean
manhattan
k
Calinski-Harabasz
Higher is better
euclidean
manhattan
k


Distance-Metric Agreement
Partition Agreement Across Distance Metrics
ARI between Euclidean and Manhattan partitions
1.0
0.8
0.6
0.4
0.2
0.0
2
3
4
5
6
7
8
k
Low sensitivity threshold
Moderate sensitivity threshold

Final Algorithm Comparison